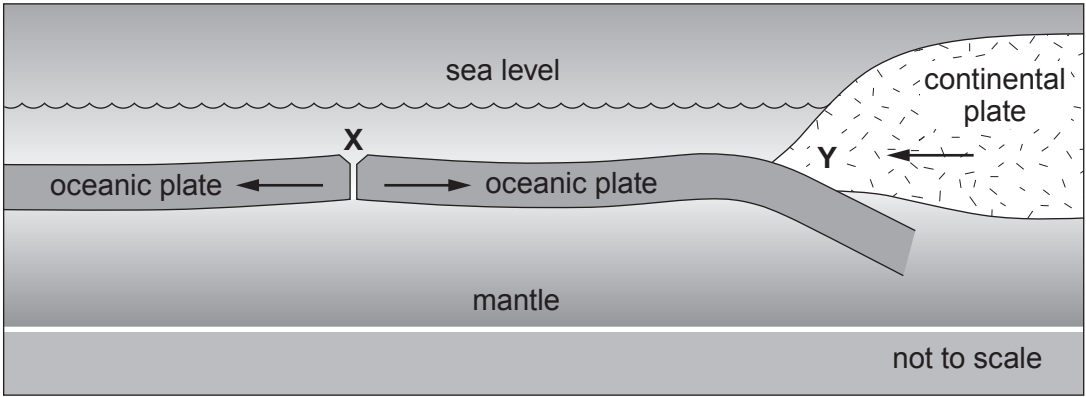


4. The Earth’s crust is broken up into tectonic plates.
- The diagram shows two plate boundaries **X** and **Y**.



- (a) Describe what happens as the plates move apart at plate boundary **X**. [2]

.....

.....

.....



(b) The two types of plate have different densities as shown in the table.

Type of plate	Density (g/cm ³)
continental	4.7
oceanic	5.0

At boundary **Y** the oceanic plate is pushed **underneath** the continental plate.

(i) State why the oceanic plate is pushed underneath the continental plate. [1]

.....

(ii) State what happens to the oceanic plate when it goes underneath the continental plate. [1]

.....

(iii) Underline the name of the type of plate boundary seen at **Y**. [1]

destructive constructive conservative

(c) Using satellite imaging, scientists discovered that one oceanic plate had moved 537 cm over 600 years. Calculate the mean rate of movement of the plate over this time. [2]

rate = $\frac{\text{distance}}{\text{time}}$

Rate = cm/year

7

